18-MODEL-EVAL-GATES.md — No model ships without beating the baseline

Rule R5 — Quality is a gate, not a hope. In an AI product, the scariest bug is not a crash. It is a model that quietly gets 4 % worse at skin tones and nobody notices until a bride does. This file makes quality regression mechanically impossible to merge.

1. The eval contract

Every model in `02-AI-MODEL-STACK.md` must ship with five artefacts. A model missing any one of them cannot be packaged, signed, or loaded.

Artefact	File	Purpose
Signature	`models/<name>/signature.json`	Input/output shapes, normalisation, colour space
Model card	`models/<name>/CARD.md`	Training data, intended use, known failure modes, prohibited uses
Eval report	`models/<name>/eval.json`	Metrics on the frozen eval set, per slice
Calibration	`models/<name>/calibration.json`	Temperature/isotonic parameters and the resulting ECE
Parity report	`models/<name>/parity.json`	Cross-EP agreement per `12-DETERMINISM-CONTRACT.md`

2. Eval set discipline

- **Three splits, physically separated.** `train` / `dev` / `eval`. The eval set is write-once and is never used for tuning. It lives in a separate bucket with a different access path so it cannot be touched casually.
- **Split by wedding, never by photo.** Two frames from the same ceremony in different splits is leakage that inflates every metric you care about.
- **Frozen and versioned.** `eval-v1` is a fixed manifest of file hashes. Growing the eval set creates `eval-v2` and requires re-baselining everything, in its own PR.
- **Leakage test in CI.** Assert zero hash overlap and zero wedding-ID overlap between splits. This test has caught more fake accuracy than any other single check.

3. Slices — the part most teams skip

Aggregate accuracy hides exactly the failures that damage a wedding business. Every metric is reported **per slice**, and a gate fails if any slice regresses, even when the average improves.

Slice axis	Buckets
Skin tone	Monk scale 1–10, minimum 200 faces per bucket
Lighting	Daylight, golden hour, tungsten, mixed, candlelight, flash, night
Venue	Indoor dark, indoor bright, outdoor sun, outdoor shade, tent
Culture / attire	Minimum 8 wedding traditions represented, including South Asian, East Asian, African, Western, Middle Eastern
Camera	Canon, Nikon, Sony, Fuji; 24–61 MP; ISO 100–12800
Motion	Static, walking, dancing, confetti, sparklers
Group size	1, 2, 3–10, 11–50, 50+

Fairness gate: the spread between the best and worst skin-tone bucket may not exceed **3 percentage points** on any primary metric, and it may not widen versus the previous release. This is both an ethics requirement and a product requirement.

4. Gates per model family

Model	Primary metric	Ship gate	Regression tolerance
Focus / blur	AUC vs human labels	≥ 0.97	− 0.005 absolute
Eyes closed	Recall @ 1 % FPR	≥ 0.95	− 0.01
Face recognition	TAR @ FAR 1e-4	≥ 0.98	− 0.005, and no slice below 0.95
Scene / timeline	Segment F1	≥ 0.90	− 0.01
Expression / moment	Spearman vs expert ranking	≥ 0.75	− 0.02
Culling agreement	Overlap with photographer's own cull	≥ 85 % on keepers, ≤ 2 % false rejects of hero shots	Zero tolerance on hero false rejects
White balance	Median ΔE2000 vs expert grade	≤ 2.0, p95 ≤ 4.0	• 0.2
Exposure	Median EV error	≤ 0.15 EV, p95 ≤ 0.4	• 0.03
Style match	Blind A/B preference vs photographer's own edit	≥ 45 % preferred or tied	− 3 pp
Retouch	Texture-retention index + identity-drift = 0	TRI ≥ 0.85, landmark drift ≤ 0.5 px	Zero tolerance on identity drift
Denoise	LPIPS + detail-retention	LPIPS ≤ baseline, no detail loss ≥ 5 %	0
Masking	Boundary IoU	≥ 0.92 on hair and veil	− 0.01

5. Calibration gate

Autonomy bands (≥ 0.98 auto, 0.90–0.98 zero-touch, 0.75–0.90 suggest, < 0.75 review) are a lie unless confidence is calibrated.

- Expected Calibration Error **≤ 0.03** on 15 bins, per model, per slice.
- Reliability diagram committed as a PNG in the model folder for human review.
- **Empirical band check:** photos auto-applied at ≥ 0.98 must show a measured human-override rate of ≤ 2 % on the eval set. If overrides exceed that, the band moves — the threshold is derived from evidence, not chosen.

- Recalibrate on every release, and after any change to preprocessing, quantisation, or EP.

6. Human-in-the-loop panels

Some things cannot be measured by a metric, so they are measured by people, properly:

- **Blind A/B**, randomised order, at least 3 evaluators, at least 200 pairs; report preference with a confidence interval, not a raw percentage.
- **Artefact hunt** — `QAIQ` inspects 50 images at 100 % zoom looking for halos, banding, waxy skin, colour fringing, mask edges. Findings become fixtures.
- **Full-gallery review** — one complete 900-image gallery viewed as a client would view it, judging consistency rather than individual frames.
- Panels are required before any release, and their raw results are stored, not summarised away.

7. The `model-gate` CI job

```
1. Verify eval set hashes match the frozen manifest        ->
fail on mismatch
2. Run inference on eval-v1 for CPU reference              ->
eval.json
3. Compute metrics overall + every slice                   ->
compare to baseline.json
4. Compute ECE and reliability diagram                     ->
calibration.json
5. Run cross-EP parity                                     ->
parity.json
6. Check fairness spread and no-slice-regression rule
7. Check inference latency and memory against 17-RESOURCE-BUD
GETS.md
8. Emit a Markdown summary table into the PR
```

A model merge is blocked unless steps 3, 5, 6 and 7 all pass. Overriding a gate requires an ADR **and** a named accountable role — never a comment saying "will fix later".

8. Drift monitoring in production

- Track the **human override rate** per stage, per model version. A rise of more than 20 % relative over a week is a drift alarm.
- Track the distribution of input features (ISO, colour temperature, face size) versus training distribution; flag out-of-distribution weddings and lower autonomy automatically for them.
- Every override the photographer makes is a free label. Route them into the dataset pipeline in `07-DATASET-AND-TRAINING.md` — with consent, locally by default.
- Publish a per-release quality changelog so a photographer can see exactly what changed and why their galleries might look different.

9. Acceptance criteria

- [ ] Every model has all five artefacts; packaging refuses otherwise.
- [ ] Leakage test green (no hash and no wedding-ID overlap).
- [ ] `model-gate` job blocks merges on regression, fairness spread, parity, latency and memory.
- [ ] ECE ≤ 0.03 for every shipped model, and band thresholds are evidence-derived.
- [ ] Blind A/B results stored for the current release.
- [ ] Drift dashboard live with override-rate alarms.

10. Claude Code prompt

> Act as `MLL` (ML Lead — Vision). Implement the `model-gate` pipeline exactly as specified in `18-MODEL-EVAL-GATES.md`: the frozen eval manifest, the leakage test, per-slice metric computation, the fairness spread check, calibration with reliability diagrams, and the PR summary table. Wire packaging so a model

> without all five artefacts cannot be signed. Then switch hats to `QAIQ` and define the blind A/B protocol and the artefact-hunt checklist as runnable scripts. Report the gate output for one existing model as a table. Never lower a gate threshold to make a model pass — report the failure and the affected slices instead.